Stop Interpolation Topologically

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Agenda

Introduction

Persistent Homology

Triangulation of Manifolds

Persistence Stability

Natural Neighbor Interpolation

The Collapsing Sequence

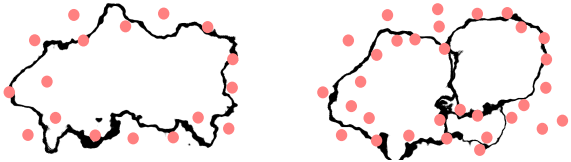
Hypothesis Testing

Summary

Introduction

Topology matters
Guess the manifold

Taking into account the topology of data is beneficial for interpolation. Data lies on some compact topological manifold, which is given by the mutual arrangement of a dataset.



Hypothesis

Two main assumptions

Hypothesis 1: Manifold assumption

A point set X lies on a *topological* or even *smooth manifold* $\mathbb{M}$, having a family of continuous/smooth *coordinate systems* to describe it, with $\dim \mathbb{M} \ll \dim X$.

Hypothesis 2: Meaning of shape

The topology of the sublevel sets of a manifold $\mathbb{M}$ underlying a data set can be used to distinguish data up to some equivalence relation.

Persistent Homology

An outline to persistent homology

A brief description

Persistent homology is the study of k-**dimensional abelian groups** attached to a topological space. They are studied algebraically and behave *similar* to the topological space itself under change revealing properties of the original space they are attached to by their *invariants*.

The connection of abelian groups by a boundary homomorphism allows to investigate all k-dimensions at once. The induced object is a chain complex.

Chain complex

How does it work?

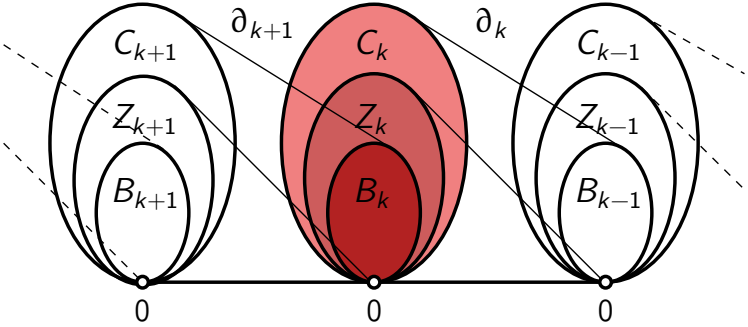
Homology theory is the study of properties of an object X, such as a set of points, by means of homological (commutative) algebra. One assigns to X a sequence of abelian groups/modules $C_0, C_1, \cdots$ which are connected by homomorphisms $\partial_{k+1} : C_{k+1} \longrightarrow C_k$ such that $\mathrm{im}\partial_{k+1} \subseteq \ker \partial_k$. Such a structure is called **chain complex**.

$$0 \xrightarrow{\partial_{k+1}} C_k \xrightarrow{\partial_k} C_{k-1} \xrightarrow{\partial_{k-1}} C_{k-2} \xrightarrow{\partial_{k-2}} \cdots \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} 0$$

Cycle and boundary groups

Putting them into context

Illustration of the chain complex



Following Zomorodian, Edelsbrunner and many more ...

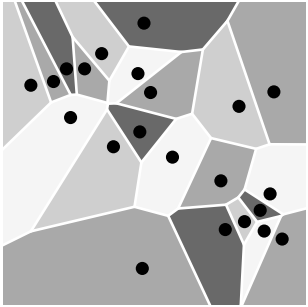
Triangulation of Manifolds

Persistence Stability

Natural Neighbor Interpolation

Step 1: Voronoi diagram

Natural neighbor interpolation example

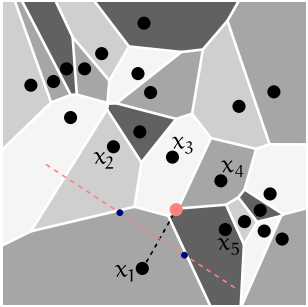


The **Voronoi Diagram** is spanned over the point cloud embedded into some Euclidean space. Technically one does not calculate the Voronoi diagram but the dual **Delaunay triangulation**.

$$\text{Vordgm} = \{x \in X \mid d(x, \lambda_i) \leq d(x, \lambda_j)\} \qquad (1)$$

Step 2: Add a point to the convex hull

We need to compute the weights per region

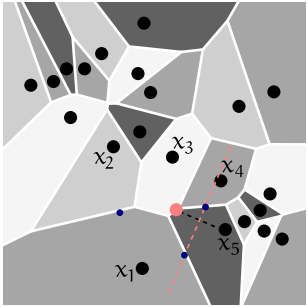


We insert a point into the convex hull of the points $X = \{x_1, \cdots, x_n\}$. The selected position of the new point follows an uniform distribution. The point is added to the set forming the convex hull:

$$\text{conv}X = \left\{ \sum_{i=1}^{n} \lambda_i x_i \;\middle|\; x_i \in X, \sum_{i=1}^{n} \lambda_i = 1, \lambda_i \geq 0 \right\}.$$

Step 3: Iterative construction of Voronoi regions

Construction of the new regions



Iteratively, the points of the Voronoi region are connected with a straight line. The perpendicular that bisects the line provides the two intersections to the neighboring Voronoi regions.

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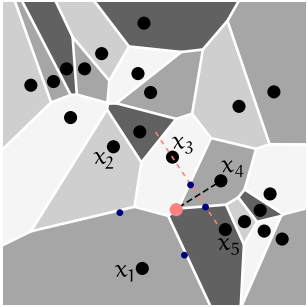

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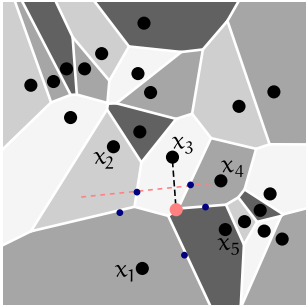

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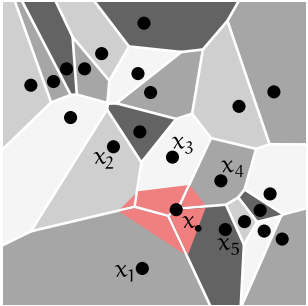
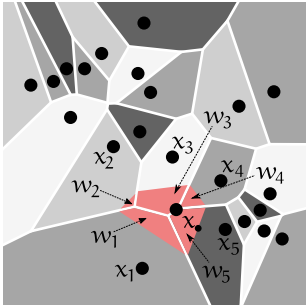


The weights w_i are the *stolen* regions of the new Voronoi region of the added point from all other Voronoi regions.

They can be calculated during construction.


w_3
x_3
x_2
x_4
w_4
w_2
x
w_1
x_5
x_1
w_5

The Collapsing Sequence

Hypothesis Testing

Summary

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